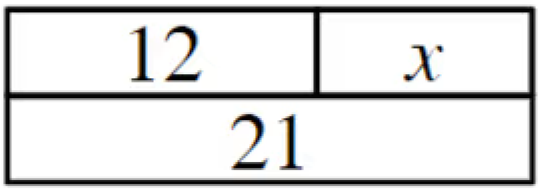
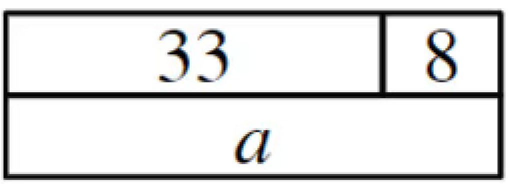
Solving simple linear equations with an additive step

1 Which equations are represented by this bar model? Tick **3** correct answers



- [x] $12 + x = 21$
- [ ] $21 + x = 12$
- [x] $21 - 12 = x$
- [ ] $21 + 12 = x$
- [x] $12 = 21 - x$

2 Which equations are represented by this bar model? Tick **2** correct answers



- [ ] $a + 8 = 33$
- [ ] $33 - 8 = a$
- [x] $a = 8 + 33$
- [x] $33 = a - 8$
- [ ] $8 = 33 - a$

3 Match each value with its additive inverse. Write the correct letter in each box

a	3
b	-3
c	$\frac{1}{3}$
d	$-\frac{1}{3}$
e	0.3
f	$\frac{10}{3}$

b	3
e	-0.3
c	$-\frac{1}{3}$
f	$-\frac{10}{3}$
d	$\frac{1}{3}$
a	-3

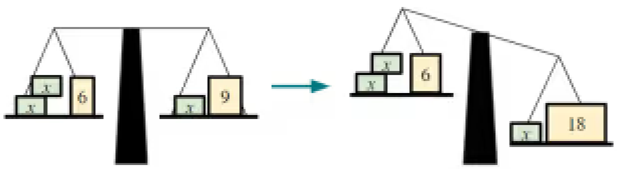
4 Which of these are zero pairs? Tick **2** correct answers

- [x] 5 and (–5)
- [ ] (–3) and (–3)
- [ ] 1 and (–0.1)
- [ ] $\frac{1}{2}$ and 2
- [x] $\frac{2}{5}$ and (-0.4)

5 What is the value of the expression $2(3x + 5)$ when $x = -3$? Fill in the blank

-8

6 The diagram shows a balanced scale on the left. When an operation is performed to the RHS, it is not balanced. What operation must be performed to the LHS to maintain equality? Tick **1** correct answer



- [ ] $\times 2$
- [ ] $\times 3$
- [x] $+ 9$
- [ ] $+ 12$
- [ ] $+ 18$